# Funkcje analityczne

Baza danych: PostgreSQL

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# Rollup

| | 123 department_id | 123 sum |
|---|---|---|
| 1 | [NULL] | 691,400 |
| 2 | 20 | 19,000 |
| 3 | 50 | 156,400 |
| 4 | 30 | 24,900 |
| 5 | 110 | 20,300 |
| 6 | 70 | 10,000 |
| 7 | 100 | 51,600 |
| 8 | [NULL] | 7,000 |
| 9 | 80 | 304,500 |
| 10 | 90 | 58,000 |
| 11 | 60 | 28,800 |
| 12 | 10 | 4,400 |
| 13 | 40 | 6,500 |

```
select
    department_id,
    sum(salary)
from employees
group by rollup(department_id)
```

# Rollup – wiele grup

| | department_id | job_id | sum |
|---|---|---|---|
| 1 | [NULL] | [NULL] | 48,300 |
| 2 | 20 | MK_REP | 6,000 |
| 3 | 10 | AD_ASST | 4,400 |
| 4 | 30 | PU_CLERK | 13,900 |
| 5 | 30 | PU_MAN | 11,000 |
| 6 | 20 | MK_MAN | 13,000 |
| 7 | 20 | [NULL] | 19,000 |
| 8 | 30 | [NULL] | 24,900 |
| 9 | 10 | [NULL] | 4,400 |

```
select
    department_id,
    job_id,
    sum(salary)  sum
from employees
where department_id in (10,20,30)
group by rollup(department_id, job_id)
```

# Cube

| 123 department_id | 123 sum |
|---|---|
| [NULL] | 48,300 |
| 20 | 19,000 |
| 30 | 24,900 |
| 10 | 4,400 |

```
select
    department_id,
    sum(salary)  sum
from employees
where department_id in (10,20,30)
group by cube(department_id)
```

# Cube – wiele grup

| 123 department_id | A-Z job_id | 123 sum |
|---|---|---|
| [NULL] | [NULL] | 48,300 |
| 20 | MK_REP | 6,000 |
| 10 | AD_ASST | 4,400 |
| 30 | PU_CLERK | 13,900 |
| 30 | PU_MAN | 11,000 |
| 20 | MK_MAN | 13,000 |
| 20 | [NULL] | 19,000 |
| 30 | [NULL] | 24,900 |
| 10 | [NULL] | 4,400 |
| [NULL] | PU_CLERK | 13,900 |
| [NULL] | AD_ASST | 4,400 |
| [NULL] | MK_REP | 6,000 |
| [NULL] | MK_MAN | 13,000 |
| [NULL] | PU_MAN | 11,000 |

```
select
    department_id,
    job_id,
    sum(salary) sum
from employees
where department_id in (10,20,30)
group by cube(department_id, job_id)
```

# Grouping

| 123 department_id | 123 sum | 123 grouping |
|---|---|---|
| [NULL] | 48,300 | 1 |
| 20 | 19,000 | 0 |
| 30 | 24,900 | 0 |
| 10 | 4,400 | 0 |

```
select
    department_id,
    sum(salary)  sum,
    grouping(department_id)  grouping
from employees
where department_id in (10,20,30)
group by rollup(department_id)
```

# Grouping – wiele grup

| department_id | sum | grouping_dept | grouping_job |
|---|---|---|---|
| [NULL] | 48,300 | 1 | 1 |
| 20 | 6,000 | 0 | 0 |
| 10 | 4,400 | 0 | 0 |
| 30 | 13,900 | 0 | 0 |
| 30 | 11,000 | 0 | 0 |
| 20 | 13,000 | 0 | 0 |
| 20 | 19,000 | 0 | 1 |
| 30 | 24,900 | 0 | 1 |
| 10 | 4,400 | 0 | 1 |

```
select
    department_id,
    sum(salary) sum,
    grouping(department_id) grouping_dept,
    grouping(job_id) grouping_job
from employees
where department_id in (10,20,30)
group by rollup(department_id, job_id)
```

# Grouping – null interpretation

| department_id | job_id | sum | grouping_dept | grouping_job |
|---|---|---|---|---|
| all department | all job_id | 48,300 | 1 | 1 |
| 20 | MK_REP | 6,000 | 0 | 0 |
| 10 | AD_ASST | 4,400 | 0 | 0 |
| 30 | PU_CLERK | 13,900 | 0 | 0 |
| 30 | PU_MAN | 11,000 | 0 | 0 |
| 20 | MK_MAN | 13,000 | 0 | 0 |
| 20 | all job_id | 19,000 | 0 | 1 |
| 30 | all job_id | 24,900 | 0 | 1 |
| 10 | all job_id | 4,400 | 0 | 1 |

```
select
            case
                        when GROUPING(department_id)=1 then 'all department'
                        else CAST(department_id as varchar) end as department_id,
            case
                        when GROUPING(job_id)=1 then 'all job_id'
                        else CAST(job_id as varchar) end as job_id,
            sum(salary) sum,
            grouping(department_id) grouping_dept,
            grouping(job_id) grouping_job
from employees
where department_id in (10,20,30)
group by rollup(department_id,job_id)
```

# Grouping sets

| department_id | job_id | sum |
|---|---|---|
| 20 | MK_REP | 6,000 |
| 10 | AD_ASST | 4,400 |
| 30 | PU_CLERK | 13,900 |
| 30 | PU_MAN | 11,000 |
| 20 | MK_MAN | 13,000 |
| 20 | [NULL] | 19,000 |
| 30 | [NULL] | 24,900 |
| 10 | [NULL] | 4,400 |
| [NULL] | PU_CLERK | 13,900 |
| [NULL] | AD_ASST | 4,400 |
| [NULL] | MK_REP | 6,000 |
| [NULL] | MK_MAN | 13,000 |
| [NULL] | PU_MAN | 11,000 |

```
select
    department_id,
    job_id,
    sum(salary) sum
from employees
where department_id in (10,20,30)
group by
        grouping sets((department_id), (department_id, job_id), (job_id))
```